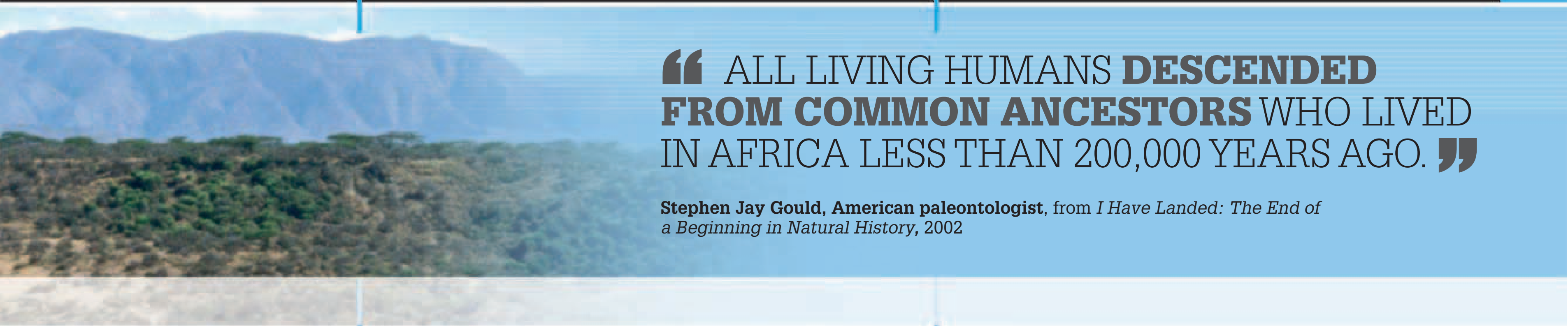


> **“** ALL LIVING HUMANS **DESCENDED FROM COMMON ANCESTORS** WHO LIVED IN AFRICA LESS THAN 200,000 YEARS AGO. **”**

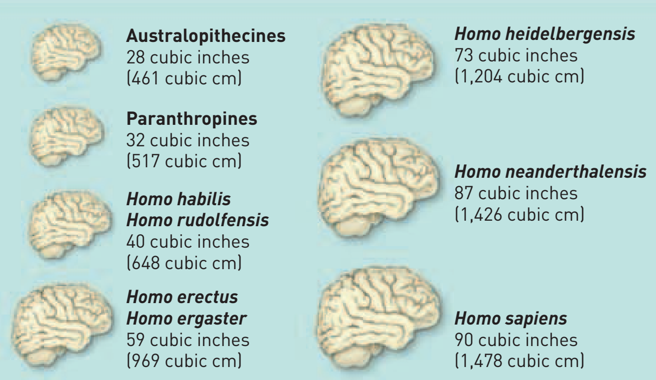
Stephen Jay Gould, American paleontologist, from *I Have Landed: The End of a Beginning in Natural History*, 2002

Hominins beyond Africa
Our earliest ancestors evolved in Africa. Possible dispersal routes from Africa are shown on this map, with dates referring to the earliest fossils known from each region.

to life in the African savanna. The climate was cooler and environments were more seasonal, with significant variation in food resources over the course of a year. Fewer edible plants meant that hominins would have had to rely more on harder-to-find and fiercely competed-for **animal protein** for food. They needed to move over greater distances and **work together** to share resources and information to **survive** in these regions.

ACHEULEAN HANDAXES made by *Homo ergaster* and *H. erectus* were produced across most of Africa and Eurasia, and demonstrate the ability to learn **complex skills** from one another and pass them down over generations. To make these tools, knappers had to think several steps ahead in order to select a suitable stone and to prepare and place each strike. Handaxes were used for a wide range of activities, including butchery, but they might also have been important for **personal or group identity**, demonstrating their makers' strength and skill.

While Homo Erectus continued to thrive in Asia, ***Homo antecessor*** had appeared as far west as northern Spain and Italy by 1.2 MYA. Marks on their bones at the site of **Atapuerca** in Spain suggest they practiced **cannibalism**. However, these early colonists may not have thrived in these unfamiliar landscapes, as very few sites are known. By 600,000 years ago, a new hominin species, ***Homo heidelbergensis***, had spread much more widely across Europe. *H. heidelbergensis* seems to have been a good hunter, or at least a proficient scavenger.



HOMININ BRAIN SIZES

Humans have a disproportionately large brain for a primate of their size, but archaeologists disagree about how and why this expansion happened. Switching to fatty and calorific foods such as bone marrow and meat may have "powered" brain growth, and also demanded more complex tools and effective hunting and foraging skills. Social skills were also a part of this process, as increasing group cooperation and pair-bonding were necessary to sustain the longer periods of childhood that infants needed for their larger brains to develop.

Burying the dead
Neanderthals often disposed of their dead with care. Some were buried in graves, as here at Kebara Cave in Israel, which dates to 60,000 BCE.

BY AROUND 350,000 YEARS AGO, while *Homo erectus* continued to hold sway over eastern Asia, *Homo heidelbergensis* in Europe and Western Asia had evolved into ***Homo neanderthalensis***.

Neanderthals were **stockier and stronger** than modern humans, and their brains were as large or even larger, although shaped slightly differently. Neanderthals were almost certainly very accomplished hunters. They were also **highly skilled** at making stone tools and heavy thrusting spears with which they tackled even large and dangerous animal prey, such as horses and bison.

However, despite burying their dead—which may have indicated ceremonial practices or belief in an afterlife—Neanderthals do not seem to have created more than the most **limited art** or used any symbols, as all modern humans do. Whether or not they spoke in a similar way to modern humans is also difficult to establish. Although their throat and voice-box anatomy suggests that a **Neanderthal language** may have been limited compared to that of humans, they must have communicated in some fashion, perhaps by combining a **less complex** form of vocalization with expressive miming.

200,000
THE NUMBER OF YEARS THE **NEANDERTHAL DOMINATED** EUROPE AND WESTERN ASIA

1.65–1 MYA Finds of ***Homo erectus*** from Java dated to as early as 1.5 mya, but more likely to date from around 1 mya

1.6 MYA Earliest known artifacts in China thought to have been made by *Homo erectus* although fossils from region currently date to only 0.8 mya

1.5 MYA "Turkana boy" is an almost **complete skeleton** of an adolescent ***Homo ergaster*** from Tanzania

1.5–1.4 MYA Evidence of **fire** at sites in South Africa, but may be natural

1.2 MYA Appearance of **first Europeans** – *Homo antecessor*

0.79 MYA First reliable evidence for **control of fire** at Gesher Benot Ya'aqov, Israel

0.78 MYA Earth's **magnetic field** assumes current polarity

0.6 MYA ***Homo heidelbergensis*** now widespread

0.4 MYA Distinctive **Neanderthal anatomy** appears across Europe

0.4 MYA Fire-hardened wooden spears in use

0.3 MYA Evidence of prepared cores used to make **multiple-component tools**

0.3 MYA Modern human skeletal traits appear in African *Homo heidelbergensis*

0.28 MYA Incised pebble from Berekhat Ram, Israel could be **first art**

0.28 MYA First evidence for use of **natural pigments**

0.2 MYA Mitochondrial Eve is the **last common ancestor** of all humans

0.186–0.127 MYA Neanderthals engage in **communal hunting** and mass kills

0.16 MYA *Homo sapiens idaltu*; skull has some primitive features, but **shares distinctive characteristics** with modern humans